

Student Performance Analysis and Academic Insights

Project Report

1. Executive Summary

This project analyses a dataset of 395 secondary-school students from two Portuguese schools, exploring factors associated with their final grade (G3, on a 0–20 scale). The analysis follows a four-level framework — Observations, Insights, Hypotheses, and Recommendations — to ensure that factual findings are clearly separated from interpretations and suggested actions.

The strongest non-grade variable associated with lower final grades is the number of prior course failures, followed by maternal education level and weekly study time. Notably, absences show near-zero linear correlation with G3 in this dataset, contrary to common expectation. A machine learning early-warning model, built using only pre-year background factors (excluding period grades G1 and G2 to avoid target leakage), achieved a cross-validated R^2 of 0.293 — indicating that background factors alone have moderate predictive value.

The top recommendation is to flag students with one or more prior failures for structured academic support at the start of the year, as this is the most consistently observed non-grade indicator of lower final performance.

2. Dataset Overview

Property	Value
Rows (students)	395
Columns (variables)	33
Schools	GP (n = 349), MS (n = 46)
Sex	F = 208, M = 187
Target variable	G3 — final grade on a 0–20 scale

Variable taxonomy:

Group	Variables
Demographic	school, sex, age, address
Family	famsize, Pstatus, Medu, Fedu, Mjob, Fjob, guardian, famsup, famrel
Academic	failures, studytime, traveltime, schoolsup, paid, reason, higher
Lifestyle & Social	activities, nursery, internet, romantic, freetime, goout, Dalc, Walc
Health	health
Other	absences
Grade outcomes	G1 (first period), G2 (second period), G3 (final — target)

Ordinal encoding keys:

- `studytime`: 1 = < 2h/wk, 2 = 2–5h/wk, 3 = 5–10h/wk, 4 = > 10h/wk
- `Medu` / `Fedu`: 0 = None, 1 = Primary (4th grade), 2 = Primary (9th grade), 3 = Secondary, 4 = Higher Education
- `failures`: number of past class failures (3 means ≥ 3)

3. Data Quality Assessment

Check	Result
Missing values	None (0 across all 33 columns)
Duplicate rows	None
Dataset modifications	None — original CSV untouched

Two distributional observations are noted:

1. **G3 = 0 pattern:** 38 students have a recorded final grade of zero. All 38 have non-zero G1 (first-period grade), making this a pattern distinct from missing data. The cause is not documented in the dataset and no assumption is made here.
2. **Right-skewed absences:** `absences` ranges from 0 to 75 with a mean of 5.71 and a median of 4. The distribution is heavily right-skewed; 75% of students have 8 or fewer absences. Five students have more than 30 absences.

No rows or values were modified. Both observations are carried through the analysis with appropriate qualifications.

4. Exploratory Data Analysis

Chart 1 — Distribution of Final Grade (G3)

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G3 is roughly bell-shaped, centred around a mean of 10.42 and a median of 11. A distinct spike at $G3 = 0$ represents 38 students whose final grade is recorded as zero. The bulk of the main distribution falls between approximately 7 and 16. The $G3 = 0$ spike introduces a slight left tail relative to the otherwise symmetric distribution.

Chart 2 — G3 by Number of Prior Failures

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Failures	n	Median G3
0	312	11.0
1	50	9.0
2	17	8.0
3	16	7.0

The median G3 declines with each increase in prior failure count. The interquartile range widens for students with more failures. Groups with 2 or 3 prior failures are small ($n = 17$ and $n = 16$) and are interpreted with caution.

Chart 3 — G3 by Weekly Study Time

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Study Time	n	Median G3
< 2h/wk	105	10.0
2–5h/wk	198	11.0
5–10h/wk	65	12.0
> 10h/wk	27	12.0

Median G3 is higher for students reporting more weekly study time. However, within-group variation is wide at all levels, and the Pearson correlation between `studytime` and G3 is only 0.098 — a weak association.

Chart 4 — Mean G3 by Maternal Education Level

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Medu	Level	n	Mean G3
0	None	3	13.00 (n too small)
1	Primary (4th grade)	59	8.68
2	Primary (9th grade)	103	9.73
3	Secondary	99	10.30
4	Higher Education	131	11.76

Among the well-populated groups (Medu 1–4), mean G3 increases progressively from 8.68 at level 1 to 11.76 at level 4, a difference of approximately 3 grade points. The Medu = 0 group (n = 3) is too small to interpret. The Pearson correlation between Medu and G3 is 0.217.

Chart 5 — Absences vs. Final Grade (G3)

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The scatter plot shows a nearly flat regression line (Pearson $r \approx 0.034$) — indicating very little linear association between absences and G3 in this dataset. Variance is high across all absence levels. Five students with absences > 30 are highlighted; their G3 values range from 8 to 11.

Chart 6 — Correlation Heatmap (Reference)

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Variable	Pearson r with G3
G2	0.905
G1	0.801
Medu	0.217
Fedu	0.152
studytime	0.098
famrel	0.051

Variable	Pearson r with G3
absences	0.034
failures	−0.360
age	−0.162
goout	−0.133
traveltime	−0.117

G1 and G2 are shown here for reference only and are excluded from the ML model (see Section 8 for the leakage explanation).

5. Observations and Insights

Note: The following insights describe associations in this dataset. They do not establish cause and effect.

Key Observations

- G3 = 0 pattern:** 38 students have G3 = 0. All have non-zero G1. The cause is not documented and is treated as an observation only.
- Prior failures:** Median G3 declines monotonically as failures increases from 0 (median = 11) to 3 (median = 7). Pearson $r = -0.360$, the strongest linear association among non-grade numeric variables.
- Study time:** Median G3 increases from 10 (<2h/wk) to 12 (≥5h/wk). Pearson $r = 0.098$. Within-group variation is wide.
- Maternal education:** Mean G3 increases from 8.68 (Medu = 1) to 11.76 (Medu = 4). Pearson $r = 0.217$. Medu = 0 group (n = 3) excluded.
- Absences:** Pearson $r = 0.034$ — no meaningful linear association with G3 in this dataset. The scatter plot confirms a near-flat trend with high variance.
- Grade associations:** G2 ($r = 0.905$) and G1 ($r = 0.801$) have by far the strongest linear associations with G3. These are excluded from the ML model.

Insights

- Prior failures are the non-grade variable most consistently associated with lower G3.** Each increase in failure count corresponds to a lower median G3 ($r = -0.360$), and the pattern holds across all four groups.

2. **Higher maternal education level is associated with higher mean G3** across the well-populated groups (Medu 1–4). The difference between the lowest and highest well-populated groups is approximately 3 grade points.
3. **Greater weekly study time is associated with higher G3**, but the relationship is weak ($r = 0.098$) and wide within-group variation means study time alone does not reliably distinguish performance levels.
4. **Absences show no meaningful linear association with G3 in this dataset** ($r = 0.034$). This is a dataset-specific finding and may not hold in other populations.
5. **The 38 students with G3 = 0 form a pattern distinct from the main distribution.** All had non-zero G1, and the sensitivity analysis (Section 8) shows they materially affect model metrics.

6. Hypotheses

The following are plausible explanations requiring additional evidence to test.

1. **Compounding academic difficulty:** Students who have previously failed a course may face cumulative challenges — such as knowledge gaps or reduced engagement — that persist into the final period. Whether targeted support after a first failure changes this pattern could be tested by comparing outcomes of students who received structured intervention versus those who did not.
2. **Home learning environment:** Higher maternal education may be associated with greater access to learning resources, academic guidance, or higher expectations at home — any of which could partially account for the observed grade difference. A study collecting home-environment data would be needed to test this.
3. **Nature of G3 = 0 records:** The 38 G3 = 0 students may represent a qualitatively different outcome — such as non-submission, administrative recording, or withdrawal — rather than measured academic performance of zero. This should be investigated through institutional records.

7. Recommendations

#	Recommendation	Evidence Strength
1	Flag students with ≥ 1 prior failure for structured academic support at the start of the year. This is the most consistently observed non-grade indicator ($r = -0.360$, monotonic decline across all failure levels).	Supported by consistent observation
2	Provide study-skill support for students in the lowest study-time band ($< 2\text{h/wk}$, $n = 105$). This group shows the lowest median G3 (10). The association is modest but the factor is measurable and actionable.	Supported by moderate association
3	Use the combination of prior failures and very low study time as a composite early-warning profile, rather than relying on any single variable in isolation.	Supported by observation; indirect evidence for action
4	Note the maternal education gradient when designing student support programmes. The observed ~ 3 -point mean G3 difference between $\text{Medu} = 1$ and $\text{Medu} = 4$ is real, though its mechanism is unclear from this data alone.	Supported by observation; mechanism unclear
5	Audit the 38 $\text{G3} = 0$ records to determine their institutional meaning before using this dataset in further analysis or modelling.	Requires investigation before action

8. Machine Learning Approach

Scenario and Leakage Reasoning

Why G1 and G2 are excluded: G1 and G2 are period grades from the same academic year as G3. G2 alone has a Pearson correlation of 0.905 with G3. A model using these features would appear highly accurate but would only be applicable after those grades already exist — offering no

early-warning value. The chosen scenario predicts G3 using only information available before the academic year begins.

Features excluded: G1, G2 (target leakage), G3 (target variable) **Features included:** All remaining 30 columns **Model:** Random Forest Regressor (n_estimators = 100, random_state = 42) **Encoding:** Binary yes/no columns → 0/1; multi-category columns → one-hot (drop_first = True); ordinal integers kept as-is

Evaluation Results

Primary evaluation — 5-fold cross-validation (n = 395):

Metric	Result
CV RMSE	3.819 ± 0.238
CV R²	0.293 ± 0.040

Single 80/20 split (supplementary; used for feature importance chart):

Metric	Result
Baseline RMSE (predict training mean)	4.550
Model Test RMSE	3.921
Model Test R²	0.250

The model improves on the naive baseline (RMSE 4.550 → 3.921 on the test split; cross-validated RMSE of 3.819). The R² of 0.293 indicates that background factors alone explain approximately 29% of G3 variance. This is modest — as expected and appropriate for an early-warning model without access to period grades. A substantially higher R² without G1/G2 would be a signal to investigate for additional leakage.

Limitation: A single 80/20 split on 395 rows is sensitive to the random seed. Cross-validation is the primary reliability measure.

Feature Importances (Top 15)

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The model assigned the highest importance scores to:

Rank	Feature	Importance
1	absences	0.1926

Rank	Feature	Importance
2	failures	0.1424
3	health	0.0550
4	goout	0.0511
5	age	0.0437
6	studytime	0.0345
7	freetime	0.0318
8	Fedu	0.0317
9	traveltime	0.0316
10	Walc	0.0297

Note on feature importances: These are the model's internal weightings (mean decrease in impurity), not validated causal relationships. The high importance of `absences` in the model contrasts with its near-zero linear correlation with G3 ($r = 0.034$). This reflects that the Random Forest captures non-linear and interaction effects that Pearson correlation does not — `absences` may interact with other features in ways that help partition the data, without having a simple linear relationship with G3. `failures` ranking second is consistent with its being the strongest linear non-grade correlate.

Sensitivity Analysis — G3 = 0 Records

Dataset	CV RMSE	CV R ²
Full (n = 395)	3.819 ± 0.238	0.293 ± 0.040
Excluding G3 = 0 (n = 357)	2.911 ± 0.160	0.179 ± 0.049
Change	−0.909	−0.114

Excluding the 38 G3 = 0 rows reduces RMSE by approximately 0.91 grade points, indicating those records are disproportionately difficult to predict and inflate the overall error. The R² also drops, which reflects that the remaining distribution is more concentrated and harder to explain proportionally. This demonstrates that the G3 = 0 records materially affect both error and explained- variance metrics, and supports the recommendation to audit their institutional meaning.

9. Conclusion

This project applied structured exploratory data analysis to a dataset of 395 secondary-school students, using a four-level analytical framework to distinguish data observations from insights, hypotheses, and recommendations.

Key analytical findings:

- Prior failures ($r = -0.360$) are the strongest non-grade linear indicator of final performance in this dataset.
- Maternal education ($r = 0.217$) shows a meaningful association across well- populated groups.
- Study time shows a weak but present association ($r = 0.098$).
- Absences, despite common expectation, show near-zero linear correlation with G3 ($r = 0.034$) in this sample.
- 38 students with G3 = 0 form a distinct sub-group that materially affects both descriptive statistics and model metrics.

Machine learning: The Random Forest early-warning model achieved CV RMSE = 3.819 and CV $R^2 = 0.293$ using 30 background features, without any period grades. This represents a meaningful improvement over a naive baseline (RMSE = 4.550) and demonstrates that pre-year factors have moderate — though not strong — predictive value.

Next steps that would strengthen this work:

1. Auditing the 38 G3 = 0 records through institutional records to clarify their meaning.
2. Collecting home-environment and socioeconomic data to test the hypotheses about parental education.
3. Using a longitudinal or controlled study design to move from associations to causal claims.
4. Applying the early-warning model prospectively to measure whether intervention for flagged students improves outcomes.